

```
# Python program for arithmetic operations
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```
a = float(input("Enter first number: "))
```

```
b = float(input("Enter second number: "))
```

```
print("Addition =", a + b)
```

```
print("Subtraction =", a - b)
```

```
print("Multiplication =", a * b)
```

```
if b != 0:
```

```
    print("Division =", a / b)
```

```
    print("Modulus =", a % b)
```

```
else:
```

```
    print("Division and modulus are not possible because the second number is 0.")
```

```
print("Exponentiation =", a ** b)
```

Sample Output:

Enter first number: 10

Enter second number: 2

Addition = 12.0

Subtraction = 8.0

Multiplication = 20.0

Division = 5.0

Modulus = 0.0

Exponentiation = 100.0